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ODD END SEMESTER EXAMINATION DECEMBER-2024

Name of the Course: B.Tech.-
Aerospace Engineering

Semester: 3rd

Name of the Paper: Introduction to
Aerospace Engineering

Paper Code: TAS-302

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Describe the major milestones in the history of Indian spacecraft. Include key achievements such as the launch of India's first satellite, major missions like Mars Orbiter Mission (Mangalyaan), and their impact on space exploration.	CO1
(b)	Describe the different layers of Earth's atmosphere with a suitable diagram. Discuss their key characteristics, including altitude range, temperature variation, composition, and notable features.	CO1
(c)	Explain the classification of aircraft and describe the basic components that make up an aircraft.	CO2
Q2	(20marks)	
(a)	Describe the nomenclature of an aerofoil, explaining the key terms used in its identification and design. Additionally, explain the different wing planforms commonly used in modern aircraft, providing examples for each.	CO2, CO3
(b)	Describe the different types of aircraft motions and explain the role of different control surfaces with suitable diagrams.	CO3
(c)	Provide a detailed explanation of the following concepts, supported by appropriate diagrams where necessary: a) Centre of Pressure and its Significance b) Pressure Distribution Over a Wing Section c) Factors Influencing Lift and Drag Forces	CO3
Q3	(20marks)	
(a)	Describe the basic components of an aircraft truss structure and explain their functions. Include a suitable diagram to illustrate the layout and design of the truss structure.	CO6
(b)	Define the monocoque structure and explain the concept of an aircraft monocoque skeleton. Discuss its applications in aircraft design and provide an example of an aircraft that utilizes this structural design.	CO2, CO6
(c)	Classify aerospace materials using a flowchart and provide an explanation for each category. Additionally, identify and list the specific materials commonly used in various components of an aircraft.	CO6
Q4	(20marks)	
(a)	Describe the fundamental differences between air-breathing and non-air-breathing engines with suitable examples.	CO4
(b)	Summarize the operating principle of a turbofan engine and identify its key components. Discuss the role of each component in the engine's overall performance.	CO4

(c)	Outline the main components of a solid propellant rocket engine and explain the propellant configuration typically used.	CO4
Q5	(20marks)	
(a)	Explain the primary functions of the aircraft's hydraulic and pneumatic systems. Discuss how these systems work together to support various aircraft operations and ensure overall performance.	CO5
(b)	What are aircraft instruments, and why are they essential for safe and efficient aircraft operation? List and briefly describe the different types of instruments used in aircraft.	CO5
(c)	Discuss the various satellite navigation systems developed by countries around the world. How does India's NavIC system contribute to regional navigation services?	CO5